

Module 4: Basic Time & Date Functions

4.1 Concept

Time and date functions in DAX allow you to perform **time-based analysis**.
In our dataset, the **OrderDate** and **ShipDate** columns are ideal for:

- Calculating delivery times
- Grouping by year/month
- Performing year-over-year comparisons

4.2 Key Functions Overview

Function	What it Does	Example (on dataset)
DATE(year, month, day)	Creates a date	DATE(2025, 9, 17)
YEAR(date)	Extracts year	YEAR(Sales[OrderDate])
MONTH(date)	Extracts month	MONTH(Sales[OrderDate])
DAY(date)	Extracts day	DAY(Sales[OrderDate])
DATEDIFF(start, end, interval)	Returns difference in interval	DATEDIFF(OrderDate, ShipDate, DAY)

Function	What it Does	Example (on dataset)
DATEADD(date, number, interval)	Shifts date by interval	DATEADD(Sales[OrderDate], -1, YEAR)
TODAY()	Returns current date	TODAY()
NOW()	Returns current date & time	NOW()

4.3 Practical Examples

Example 1: Calculate Delivery Days

Days to Ship :=

DATEDIFF(Sales[OrderDate], Sales[ShipDate], DAY)

- ◆ Returns number of days it took to ship each order.

Example 2: Extract Order Year & Month

Order Year := YEAR(Sales[OrderDate])

Order Month := MONTH(Sales[OrderDate])

- ◆ Useful for trend analysis or time-based grouping.

Example 3: Sales Last Year

```
Sales Last Year :=  
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    DATEADD(Sales[OrderDate], -1, YEAR)  
)
```

- ◆ Compares current year sales to previous year.

Example 4: Orders Today

```
Orders Today :=  
CALCULATE(  
    COUNT(Sales[SalesID]),  
    Sales[OrderDate] = TODAY()  
)
```

- ◆ Counts how many orders were placed today.

4.4 Exercises (Hands-On)

1. Delivery Time Flag

Create a column:

- Days to Ship <= 3 → "Fast"

- 4–7 → "Normal"
- 7 → "Slow"

2. Extract Order Year-Month

- Create a column "YYYY-MM" combining YEAR and MONTH for time-based reporting.

3. Orders This Month

- Create a measure counting orders where MONTH(OrderDate) = MONTH(TODAY()) .

4. Days Since Order

- Measure: DATEDIFF(OrderDate, TODAY(), DAY) → How many days since each order.

5. Sales Previous Month

- Measure: Sum SalesAmount shifted one month back using DATEADD .

4.5 Visualizations in Power BI

1. Line Chart

- X-axis = Order Year-Month
- Y-axis = Total Sales

2. Table

- Product | OrderDate | ShipDate | Days to Ship | Delivery Time Flag

3. Card Visual

- Orders Today / Orders This Month

4.6 Exercise Answers (For Trainers)

1. Delivery Time Flag

```
Delivery Flag :=  
SWITCH(  
    TRUE(),  
    DATEDIFF(Sales[OrderDate], Sales[ShipDate], DAY) <= 3, "Fast",  
    DATEDIFF(Sales[OrderDate], Sales[ShipDate], DAY) <= 7, "Normal",  
    "Slow"  
)
```

- "Fast" : ~10–15% of orders
- "Slow" : ~5–10% of orders

2. Order Year-Month

```
Order YYYY-MM :=  
FORMAT(Sales[OrderDate], "YYYY-MM")
```

- Example: "2024-03" for March 2024 orders

3. Orders This Month

```
Orders This Month :=  
CALCULATE(  
    COUNT(Sales[SalesID]),  
    MONTH(Sales[OrderDate]) = MONTH(TODAY())  
)
```

- Counts all orders in the current month.

4. Days Since Order

Days Since Order :=

```
DATEDIFF(Sales[OrderDate], TODAY(), DAY)
```

- Provides a dynamic count of days elapsed per order.

5. Sales Previous Month

Sales Prev Month :=

```
CALCULATE(  
    SUM(Sales[SalesAmount]),  
    DATEADD(Sales[OrderDate], -1, MONTH)  
)
```

- Sum of sales shifted one month back for comparison.

✅ By the end of Module 4, students will:

- Extract **year, month, day** from date fields.
- Calculate **delivery times** and **days since order**.

- Build **time-based measures** for trend and comparison analysis.
- Use `DATEADD` , `DATEDIFF` , `TODAY` , `NOW` for real-world reporting scenarios.